

Assignment: PII Redaction Tool

Background

Attached is Red Herring Prospectus, document with PII information.

Task

Write a script (language of your choice) that reads the ticket log and produces a redacted version, replacing all personally identifiable information (PII) with fake alternative. For example:

```
Rashi Patil: John Doe
rashhi.patil@gmail.com: john.doe@example.com
Rohan Dey: Peter Parker
rohan.dey@gmail.com: peter.parker@example.com
+91 9876543210: +91 9876543210
```

Your solution should detect and redact, at minimum:

- Full names
- Email addresses
- Phone numbers
- Company names
- Physical/mailling addresses
- Social Security Numbers (SSNs)
- Credit card numbers
- Dates of birth
- IP addresses

Deliverables

1. Source code for your redaction script.
2. The redacted output file in docx.
3. A short README (a few sentences) explaining your approach (e.g., regex-based, NER model, third-party library) and any tradeoffs or false positives/negatives you noticed.
4. **Explain the evaluation approach that you have used and prepare a evaluation report for your run - include the accuracy, precision and recall numbers in the report.**

Evaluation Criteria

- **Recall:** did you catch all instances of each PII type?
- **Precision:** did you avoid redacting things that weren't actually PII (e.g., "Order" or "Ticket" numbers, unless you choose to treat those as sensitive too; reasonable either way, just be explicit about your choice)?
- **Code quality:** readability, structure, and how you'd extend it to a new PII type.
- **Communication:** clarity of your README.

There's no single "correct" tool to use; regex, spaCy/NER, Presidio, or your own logic are all fair game.